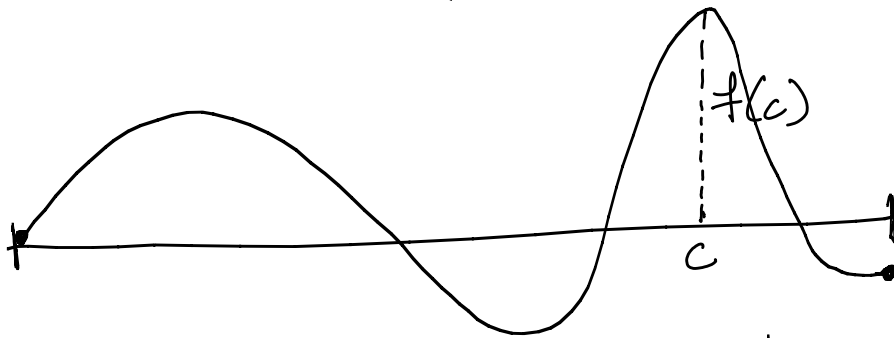


Unit 4.1 Extreme values

Note Title

2015/04/15

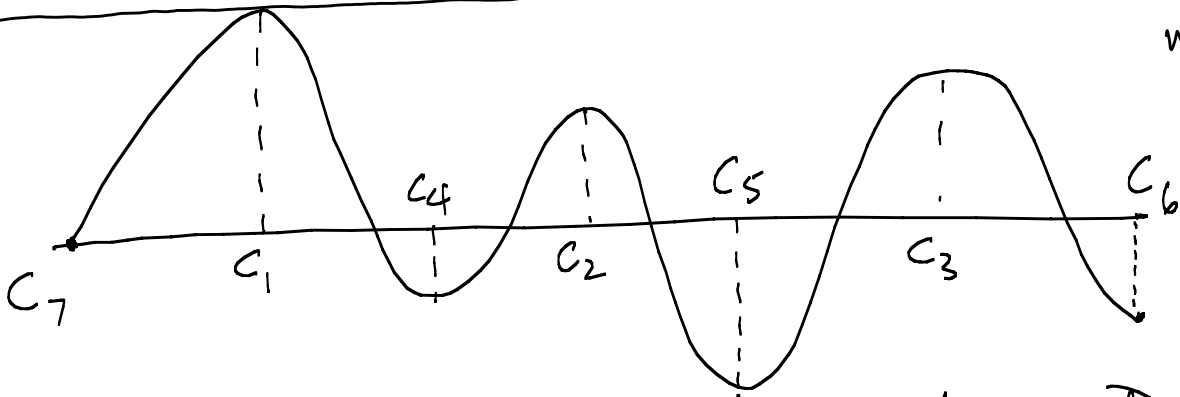
Absolute maximum / Absolute minimum



Let D be the domain of the function f .

- $f(c)$ is the absolute maximum of f on D if $f(c) \geq f(x)$ for all $x \in D$.
- $f(c)$ is the absolute minimum of f on D if $f(c) \leq f(x)$ for all $x \in D$.

Local maximum & local minimum (Also called relative max/min)



$f(c)$ is a local maximum of f on D if $f(c) \geq f(x)$ for all x values close to c .

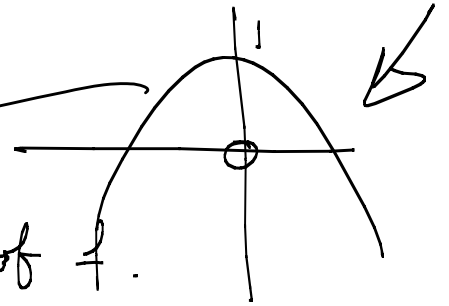
$f(c)$ is a local minimum of f on D if $f(c) \leq f(x)$ for all x values close to c .

We say and write:

- An absolute / local maximum / minimum is called an **extreme**
eg. Find the local extremes.....
local max / min.

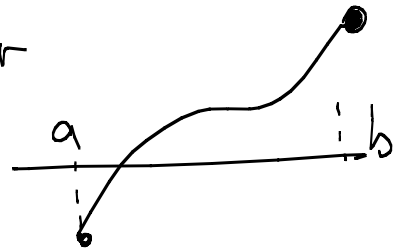
- An abs/local extreme is a **function value**

$$f(x) = -x^2 + 1$$



- $f(0) = 1$ is the absolute max of f .
or f has an abs max of 1 at $x=0$.

- An absolute max/min can occur at the **endpoints** of the interval
Abs max: $f(b)$ abs min: $f(a)$



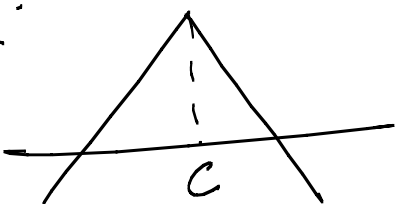
- An extreme can occur in more than one place

Critical numbers

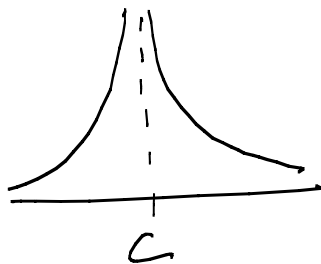
A critical number of a function f is a number c in the domain of f such that $f'(c) = 0$ or $f'(c)$ does not exist.

NB: c is in D

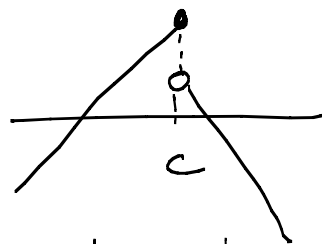
Eg.



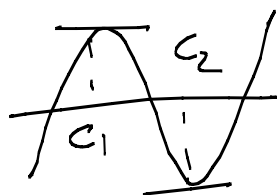
1 crit numb.



no crit numbers.



1 crit numb.



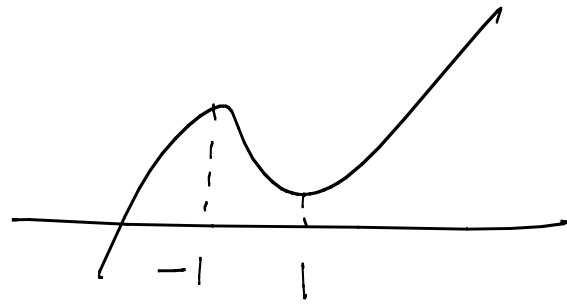
2 crit numbers

Eg. Find the critical numbers and decide whether f has an extreme at the point or not.

1. $f(x) = x^3 - 3x$

$$f'(x) = 3x^2 - 3 = 0$$

$$\Rightarrow x = \pm 1.$$



Two critical numbers : $x = 1$, $x = -1$.

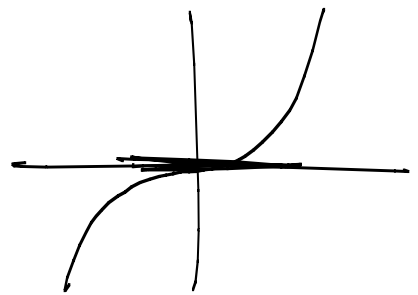
local min

local max.

2. $f(x) = x^3$

$$f'(x) = 3x^2 = 0$$

$$\Rightarrow x = 0$$

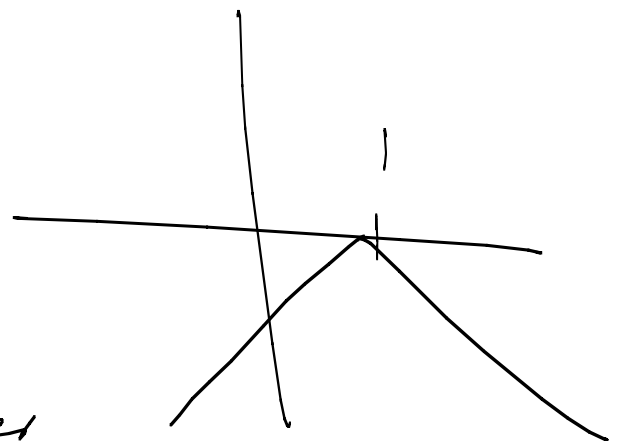


$\Rightarrow x = 0$ is a critical number.

No extreme at $x = 0$.

3. $f(x) = -|x - 1|$

$f'(x)$ does not exist at $x = 1$.



$\Rightarrow x = 1$ is a crit number

$\Rightarrow$ local and absolute max at $x = 1$.

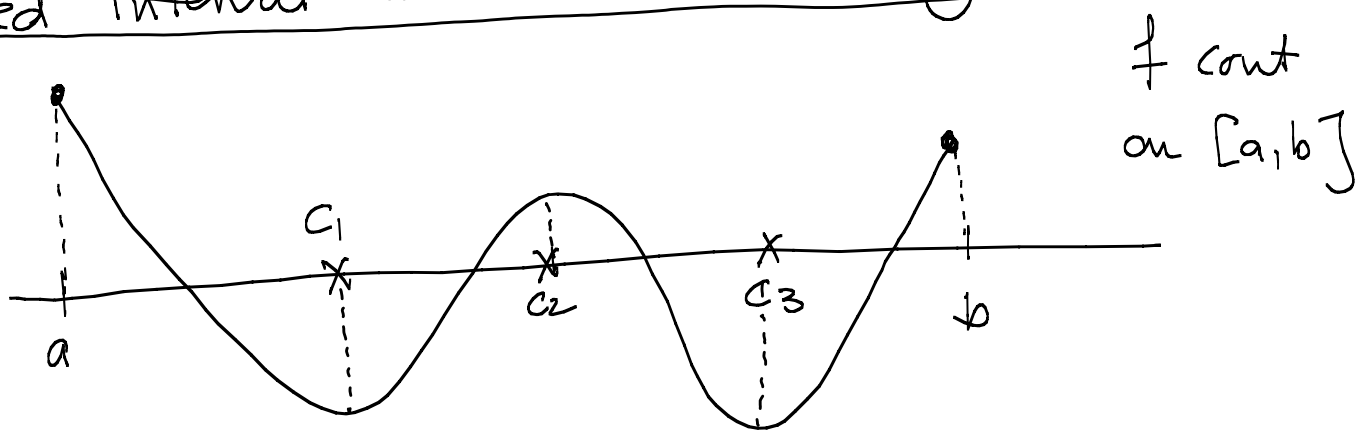
How can you be certain that the function has absolute extremes?

The following theorem answers the question

Extreme value theorem (Fermat's theorem)

If f is continuous on a closed interval then f has an absolute maximum and an absolute minimum.

Closed interval method for finding extremes:



AB:

1. Check that f is continuous on $[a, b]$
2. Find the critical numbers of f on $[a, b]$
3. Find the function values at the critical numbers and also find $f(a)$ & $f(b)$
4. The smallest number in 3 is the absolute minimum and the biggest number in 3 is the absolute maximum.

Examples: Use the Closed Interval Method to find the absolute extremes of

1. $f(x) = x^4 - 2x^2$ on $[-0.5, 2]$

1. f cont on the interval (polynomial)

2. Crit numbers:

$$4x^3 - 4x = 0$$
$$\Rightarrow 4x(x^2 - 1) = 0$$

$$\Rightarrow x = 0 \text{ or } x = -1 \text{ or } x = 1$$

not in interval

3. $f(-0.5) = \frac{1}{16} - \frac{1}{2} = -\frac{7}{16}$

$$f(0) = 0$$

$$f(1) = -1$$

$$f(2) = 8.$$

4. f has an abs max $f(2) = 8$
 f has an abs min $f(1) = -1.$

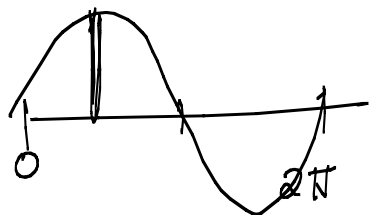
2. $f(x) = x + \cos x$ on $[0, 2\pi]$.

1. f is cont on the interval

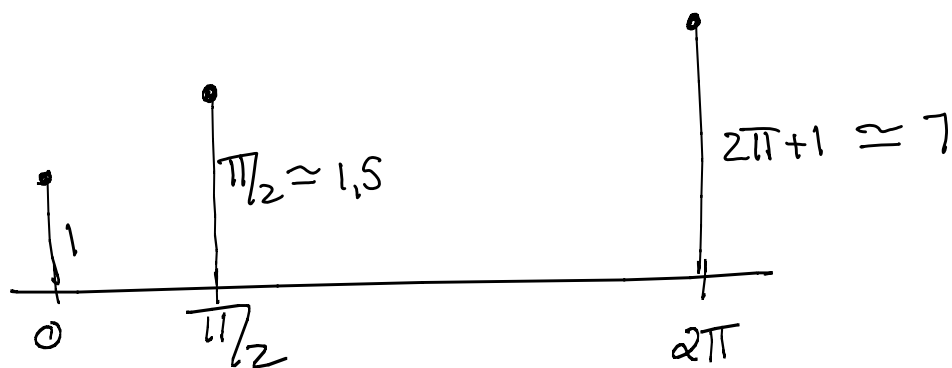
2. $f'(x) = 1 - \sin x = 0$

$$\Rightarrow \sin x = 1$$

$$\Rightarrow x = \pi/2$$



3.



4. f has an abs max $f(2\pi)=7$
 f has an abs min $f(0)=1$.

